

Button and Fold a Dress Shirt SOP

One episode closes one dress shirt and folds it. The shirt is already lying face up and spread on the fold mat when recording starts. The five snaps down the front placket are pressed closed from the collar down, the placket is smoothed straight, the shirt is turned face down on the mat, each side is folded in and its sleeve laid back behind the body, the body is folded into thirds, and the shirt finishes as one squared packet with the whole collar showing face up at the front edge.

The shirt closes with **press snaps**, not sewn buttons. A snap closes on one straight downward press and nothing else. No snap is pinched shut between the gripper fingers and no placket is pulled sideways to bring a socket onto its stud.

The order never changes: fasten, smooth, turn, fold the sides, fold in thirds. No side is folded before all five snaps are closed and the placket has been smoothed, and nothing is folded while the shirt is still face up.

The right gripper does the primary work: it lays and presses every snap, smooths the placket, and folds the right side. The left gripper supports: it pins the cloth for every press and folds the left side. Each arm works its own side of the mat and neither takes the other's jobs.

The shirt stays on the fold mat for the whole episode. Nothing is lifted into the air and turned, nothing is pushed or shoved into place, and a gripper opens nowhere but on the mat.

Setup

Complete both checklists before starting an episode.

Hardware checklist

1. Cameras are on and recording.
2. The environment camera shows the whole fold mat with the spread shirt inside it: collar, hem, both sleeves, and both cuffs in frame.
3. The overhead view shows the front placket end to end, with all five snaps readable.
4. Both arms are at home with grippers open.
5. The desk is clear of anything but the fold mat and the shirt on it.
6. The left gripper reaches the left half of the mat, the left cuff, the collar, and the placket at the center without the arm leaning out or reaching a joint limit.
7. The right gripper reaches the right half of the mat, the right cuff, the collar, and the placket at the center without the arm leaning out or reaching a joint limit.

Materials checklist

1. One **dress shirt** already lying face up and square on the fold mat, collar toward the far side and hem toward the front edge, with no part of it off a mat edge.
2. Both sleeves lie flat on the mat, angled out from the shoulders toward the front corners, cuffs flat and clear of the placket. Neither sleeve lies across the front.
3. The **over placket** rests on the **under placket** with all five snaps open. The over placket comes from the right side of the shirt, the under placket from the left.
4. The shirt closes with five **press snaps** evenly spaced down the placket from the collar to the hem. There are no sewn buttons and no buttonholes. The under placket carries the studs facing up, the over placket carries the sockets facing down.
5. Every snap closes on one straight downward press and holds. A snap that will not click closed, or that pops open on its own, is replaced before the episode.
6. The shirt has no collar snap and no cuff snaps. The five placket snaps are the whole of the fastening work.
7. The shirt is pressed flat, dry, and free of stains, tears, and open seams. It holds a fold rather than springing open.
8. The shirt still carries no fold line from a previous episode.
9. The fold mat lies flat at the center of the desk, bare apart from the shirt, and is wide enough and deep enough that the spread shirt sits inside it with clear mat all round. Its surface holds cloth still under a stroke rather than letting it skate.

Workspace layout

- **Fold mat:** flat at the center of the desk, reachable by both arms. It holds one shirt. Every press, every stroke, and every fold happens here, and it is the only place a gripper opens.
- **Left half of the mat:** the left body edge, the left sleeve, and the left cuff. The left gripper's zone.
- **Right half of the mat:** the right body edge, the right sleeve, and the right cuff. The right gripper's zone.
- **Center of the mat:** the placket and the collar. Both grippers reach it, one to pin and one to work, and neither carries anything across to the far side of it.

Vocabulary

- **Arm assignment:** the right gripper lays and presses every snap, smooths the placket, and folds the right side and right sleeve. The left gripper pins the cloth for every press and folds the left side and left sleeve. Both grippers together turn the shirt and make the thirds folds, each on its own corner. Neither gripper takes the other's jobs.
- **Front:** the side of the shirt carrying the placket. **Face up** is the front toward the ceiling, **face down** is the front against the mat.

- **Placket:** the doubled strip running down the front from the collar to the hem. The **over placket** lies on top and carries the sockets, the **under placket** lies beneath and carries the studs.
- **Snap:** one two part press fastener on the placket, a **stud** on the under placket and a **socket** on the over placket. The snaps are numbered from the collar down: snap 1 is nearest the collar, snap 5 nearest the hem.
- **Closed snap:** the two placket edges hold together when the gripper lifts off, with the edges in line, no gap, and no step between them. Nothing else counts as closed.
- **Press:** the closed gripper set straight down flat on the cloth, held still for 2 seconds, and lifted straight off. A press never slides, never rubs, and never pinches.
- **Pin:** a closed gripper pressed straight down on the cloth to hold it still. A pin never pinches the cloth and never travels.
- **Stroke:** the closed gripper pressed down on the cloth and drawn along in one continuous run, end to end, without lifting off.
- **Fold:** a gripper pinches an edge or a cuff, lifts it clear of the cloth below, carries it over, lowers it onto its target, and releases. Nothing is folded by pushing it along the mat.
- **Fold edge:** the closed folded side a fold creates, as opposed to the loose open edges.
- **Body edge:** the side seam line of the shirt, running from the shoulder to the hem.
- **Sleeve laid behind:** the sleeve lying flat on the back of the shirt, inside the side fold edge, running toward the hem end with the cuff pointing at the hem.
- **Thirds fold:** two folds that bring the hem end up toward the collar in even thirds, the second one stopping just short of the collar.
- **Packet:** the shirt once both sides and both thirds folds are done. One squared rectangle of layers with the whole collar showing face up at the front edge.
- **Settled fold:** the cloth stays put for 2 seconds after the gripper lifts clear, with no layer springing open.
- **Release point:** the fold mat. A gripper opens nowhere else.

Steps

Steps 1 to 6 run once, in order, on the one shirt. Step 7 ends the episode.

Step 1: Press the five snaps closed

Goal: all five snaps closed from the collar down, with the two placket edges in line.

- Work the snaps in order, snap 1 first and snap 5 last. A snap is closed before the next one is started.
- The **left gripper**, closed, **pins** the **under placket** just left of that snap's **stud** and holds it there until the press is finished.

- The **right gripper** pinches the **over placket** at that snap, lifts it just clear of the cloth below, lowers it so the **socket** sits on the **stud**, and releases.
- The **right gripper**, closed, **presses** straight down flat on the snap, holds for 2 seconds, and lifts straight off.
- The left gripper lifts clear only after the right gripper is off the cloth.

Lift the over placket clear and lower it onto the stud. A placket pulled sideways into line drags the whole front with it and lands the socket off its stud.

Check: the snap holds together when the gripper lifts off, the two placket edges are in line at that snap with no gap and no step between them, and no snap already closed has popped open. If the snap did not take, lift the over placket clear, lower it onto the stud again, and press again. If it still will not close after three presses, the snap is defective: stop, log it as a system issue, and discard the episode.

Step 2: Smooth the placket

Goal: the placket running straight from the collar to the hem, flat and free of rucks.

- The **left gripper**, closed, **pins** the front at the **collar band** and holds it there for the whole step.
- The **right gripper**, closed, presses down on the placket just below the collar and draws one continuous **stroke** down the placket to the **hem**, without lifting off.
- Both grippers lift clear.

The stroke follows the placket, not the panel beside it. A stroke that wanders off the placket pulls a panel out of line and can spring a snap open.

Check: all five snaps are closed, the placket runs straight from the collar to the hem with both edges in line, the front lies flat with no ruck or pucker along the placket, and the shirt is still square on the mat with nothing off a mat edge. If a snap sprang open under the stroke, close it again as in Step 1 and stroke the placket again.

Step 3: Turn the shirt face down

Goal: the shirt lying face down and square on the mat, collar toward the front edge.

- The **left gripper** pinches the **left hem corner** and the **right gripper** pinches the **right hem corner**.
- Both grippers lift the hem edge clear of the mat, carry it over the shirt toward the far side, and lay it down past the collar so the shirt turns over onto its front. Release once it is resting.
- Both grippers lift clear.
- If a sleeve has landed folded under the body or doubled on itself, the gripper on that side pinches its cuff, lifts it clear, and lays it out flat again.

The shirt turns on the mat. It is never lifted off the mat and turned in the air.

Check: the shirt lies face down and square, collar toward the front edge, hem toward the far side, both sleeves flat on the mat and angled toward the far corners, no panel folded under, no placket edge gaping at the collar or the hem, and no part of the shirt off a mat edge. If it landed skewed or bunched, take the same two corners and lay it down again.

Step 4: Fold the left side in and lay the left sleeve behind

Goal: the left side folded in on a straight fold edge, with the left sleeve flat inside it.

- The **right gripper**, closed, **pins** the back at the middle of the body and holds it there for the whole step.
- The **left gripper** pinches the **left body edge** at the middle of its length, lifts it clear, carries it toward the middle, and lays it down so the new **fold edge** runs straight from the outer edge of the collar to the hem, parallel to the placket underneath. Release once it is resting.
- The **left gripper**, closed, strokes along the fold edge from the collar end to the hem end.
- The **left gripper** pinches the **left cuff**, lifts the sleeve clear, carries it back toward the left, and lays it down so the sleeve runs toward the hem end with the cuff pointing at the hem and the whole sleeve inside the fold edge. Release once it is resting.
- The **left gripper**, closed, strokes along the sleeve from the shoulder to the cuff.
- Both grippers lift clear.

The panel is carried over and laid down. A panel dragged across the mat rucks the back under it and lands the fold edge off line.

Check: the fold edge runs straight from the collar to the hem and parallel to the placket, the sleeve lies flat inside the fold edge with the cuff pointing at the hem end, no part of the sleeve or cuff shows past the fold edge or past the hem edge, and there is no ruck along the fold. If the fold edge landed crooked or the panel came over too far, lift that edge, lay it down again, and stroke it.

Step 5: Fold the right side in and lay the right sleeve behind

Goal: the right side folded in on a straight fold edge, with the right sleeve flat inside it, so the body is now one straight sided panel.

- The **left gripper**, closed, **pins** the back at the middle of the body and holds it there for the whole step.
- The **right gripper** pinches the **right body edge** at the middle of its length, lifts it clear, carries it toward the middle, and lays it down so the new **fold edge** runs straight from the outer edge of the collar to the hem, parallel to the left fold edge. Release once it is resting.
- The **right gripper**, closed, strokes along the fold edge from the collar end to the hem end.

- The **right gripper** pinches the **right cuff**, lifts the sleeve clear, carries it back toward the right, and lays it down so the sleeve runs toward the hem end with the cuff pointing at the hem and the whole sleeve inside the fold edge. Release once it is resting.
- The **right gripper**, closed, strokes along the sleeve from the shoulder to the cuff.
- Both grippers lift clear.

Check: the two fold edges run parallel and the same distance in from the collar on each side, the right sleeve lies flat inside its fold edge with the cuff pointing at the hem end, nothing shows past either fold edge, and the body reads as one straight sided panel the width of the collar. If the two sides came out uneven, lift the right edge, lay it down again to match the left, and stroke it.

Step 6: Fold the body into thirds

Goal: the body folded into even thirds with the whole collar left showing.

- The **left gripper** pinches the **left hem corner** and the **right gripper** pinches the **right hem corner**.
- Both grippers lift the hem edge clear, carry it toward the front edge, and lay it down one third of the body's length from the collar, square to the side fold edges. Release once it is resting.
- The **right gripper**, closed, strokes along the new far fold edge from its left end to its right end.
- Both grippers pinch that **far fold edge** at its left and right corners, lift it clear, carry it toward the front edge, and lay it down **just short of the collar** so the whole collar stays showing. Release once it is resting.
- The **right gripper**, closed, strokes along the new far fold edge from its left end to its right end.
- Both grippers lift clear.

Stopping short is the whole point of the second fold. An edge laid over the collar buries it and the shirt does not present.

Check: the shirt is one squared **packet**, both fold edges run parallel to the collar, the thirds are even, the whole collar shows face up at the front edge with nothing lying over it, no sleeve or cuff shows past an edge, and no layer has sprung open. If a fold landed crooked or covered the collar, lift that fold edge back up, lay it down again, and stroke it.

Step 7: Confirm the folded shirt and end the episode

- Confirm the packet lies square on the mat, the whole collar shows face up at the front edge, both sleeves are hidden behind the body with no cuff showing past an edge, no layer has sprung open, and the mat is clear apart from the folded shirt.
- Return both arms home, clear of the mat, then stop recording.

After the episode: reset the workspace

This reset is not recorded.

1. Lift the folded shirt off the mat and open it out flat.
2. Pull the placket edges apart to open all five snaps. Check every stud and socket is undamaged and still firmly attached, and that each one still clicks closed and holds under a straight press.
3. Press the shirt flat again so no fold line from the last episode is left in the cloth. A shirt that still shows an old fold is not restaged.
4. Replace any shirt that is torn, stained, damp, has an open seam, or carries a snap that will not hold. Vary which shirt is used between episodes rather than running the same one every time.
5. Lay the shirt back face up and square on the fold mat, collar toward the far side and hem toward the front edge, both sleeves flat and angled toward the front corners with the cuffs clear of the placket, and the over placket resting on the under placket with all five snaps open.
6. Smooth the fold mat flat, confirm it has not shifted, and leave it bare apart from the shirt.
7. Wipe the desk.
8. Run both Setup checklists again.

SOP violations

These are actions that break the SOP and are reviewed side by side in the review tool.

How to record a violation in review

For each violation, record the **start timestamp**, **violation name**, and **SOP rule broken**. The visible cue is what the reviewer sees. The coaching note is for retraining and is not an annotation label.

Episode handling

Tag every violation with its timestamp and name. Keep the episode with the violation tag. Do not delete it just because a rule was broken.

Violations

Violation: Snaps worked out of order

- **Visible cue:** a snap further from the collar is closed while a snap nearer the collar is still open, or the placket is worked from the hem up.
- **SOP rule broken:** Step 1, work the snaps in order, snap 1 at the collar first and snap 5 at the hem last.

- **Coaching note:** start at the collar and count down. The top snap sets the line for the rest.

Violation: Moved on from an open snap

- **Visible cue:** the episode leaves a snap that did not take and goes on to the next snap or to the smoothing stroke, or the placket gapes at a snap position with the two edges apart.
- **SOP rule broken:** Step 1, a snap is closed before the next one is started, and Step 2 begins only with all five closed.
- **Coaching note:** watch it hold when the gripper lifts. If it springs, press it again.

Violation: Snap closed by pinching or sliding

- **Visible cue:** the gripper squeezes the two placket layers between its fingers, presses at an angle, rubs or dabs at the snap, or slides across it instead of setting down flat and holding still.
- **SOP rule broken:** Step 1, one flat press held for 2 seconds and lifted straight off, never pinched and never slid.
- **Coaching note:** down flat, count two, straight up.

Violation: Placket dragged into line

- **Visible cue:** the over placket is pushed, pulled, or shoved sideways to bring a socket onto its stud, instead of being pinched, lifted clear, and lowered onto it.
- **SOP rule broken:** Step 1, lift the over placket clear of the cloth below and lower it onto the stud.
- **Coaching note:** pick it up and put it down. Dragging takes the whole front with it.

Violation: Pressed without a pin

- **Visible cue:** the left gripper is off the cloth while a snap is pressed, it takes the pin only after the press has started, or it lifts away before the right gripper is clear, and the placket shifts under the press.
- **SOP rule broken:** Step 1, the left gripper pins the under placket beside the stud and holds it until the press is finished.
- **Coaching note:** pin first, let go last.

Violation: Arms swapped roles

- **Visible cue:** the left gripper presses a snap, smooths the placket, or folds the right side; or the right gripper folds the left side or the left sleeve. Both grippers taking a corner each in Steps 3 and 6 is not this violation.
- **SOP rule broken:** Steps 1 to 6, the right gripper snaps, smooths, and folds the right side, and the left gripper pins and folds the left side.
- **Coaching note:** left holds and takes the left. Right works and takes the right.

Violation: Stroke skipped or dabbed

- **Visible cue:** the placket, a side fold edge, a sleeve, or a thirds fold edge is left with no stroke drawn along it, or the gripper taps it in spots or presses once in the middle instead of drawing one continuous run end to end.
- **SOP rule broken:** Steps 2, 4, 5, and 6, one continuous stroke end to end without lifting off.
- **Coaching note:** one run, end to end, without lifting off.

Violation: Folded while still face up

- **Visible cue:** a side, a sleeve, or the hem is folded before the shirt has been turned, so a fold is made with the placket still showing.
- **SOP rule broken:** Step 3, the shirt is turned face down before anything is folded.
- **Coaching note:** fasten, smooth, turn. Nothing folds until the front is against the mat.

Violation: Shirt turned in the air

- **Visible cue:** the shirt is lifted clear of the mat and flipped, swung, or shaken over to its front, instead of being carried over on the mat by its two hem corners and laid down.
- **SOP rule broken:** Step 3, the shirt turns on the mat and is never lifted off it and turned in the air.
- **Coaching note:** the wrist does not have the range to turn it. Walk it over on the mat.

Violation: Fold made by dragging

- **Visible cue:** a body edge, a cuff, or the hem edge is pushed, shoved, or slid across the mat into place instead of being pinched, lifted clear, carried over, and lowered onto its target.
- **SOP rule broken:** Steps 4, 5, and 6, lift the edge clear before carrying it.
- **Coaching note:** pick it up and put it down. Nothing gets pushed.

Violation: Side fold edge off line

- **Visible cue:** a side fold edge runs crooked instead of straight from the collar to the hem, is not parallel to the other side, or the panel comes over so far that it crosses the placket line or so little that the body is left wider than the collar.
- **SOP rule broken:** Steps 4 and 5, the fold edge runs straight from the outer edge of the collar to the hem, parallel to the placket and to the other fold edge.
- **Coaching note:** line the edge up on the collar before you lay it down.

Violation: Sleeve left showing

- **Visible cue:** a cuff or part of a sleeve sticks out past a side fold edge or past the hem edge, a sleeve is left lying across the front, or a sleeve is folded but never laid flat and stays bunched under the fold.

- **SOP rule broken:** Steps 4 and 5, the whole sleeve lies flat inside the fold edge with the cuff pointing at the hem end.
- **Coaching note:** the sleeve lives behind the body. Nothing of it shows from the front.

Violation: Thirds fold buried the collar

- **Visible cue:** the second thirds fold is laid over the collar or onto the collar band, so the collar is covered or only part of it shows in the finished packet.
- **SOP rule broken:** Step 6, the second fold stops just short of the collar so the whole collar stays showing.
- **Coaching note:** stop short. The collar is what the fold presents.

Violation: Thirds uneven or out of square

- **Visible cue:** the hem edge lands well past or well short of the third line, a fold edge runs at an angle to the collar instead of parallel to it, or the finished packet's sides step out of line with each other.
- **SOP rule broken:** Step 6, the body is folded into even thirds, square to the side fold edges.
- **Coaching note:** take both corners together and keep the edge square as it comes over.

Violation: Regripped in the air

- **Visible cue:** a gripper shifts, rolls, or seats its hold again on the shirt, a panel, or a cuff without laying it down first.
- **SOP rule broken:** Steps 1 to 6, take a fresh hold only after the cloth is resting.
- **Coaching note:** put it down before you take a new hold.

Violation: Gripper released over the wrong place

- **Visible cue:** a gripper opens above the mat and the cloth falls the rest of the way, or it opens over bare desk, over open space, or off a mat edge.
- **SOP rule broken:** Steps 1 to 6, the gripper opens on the fold mat once the cloth is resting, and nowhere else.
- **Coaching note:** take it all the way down before you open the gripper.

Violation: Placement corrected by shoving

- **Visible cue:** a gripper nudges, pats, brushes, or straightens a panel, a sleeve, or the packet after releasing it, instead of lifting that edge clear and laying it down again.
- **SOP rule broken:** Steps 3 to 6, correct a bad fold by lifting that edge and laying it down again.
- **Coaching note:** lift it and lay it again, or leave it alone.

Violation: Shirt dropped, collided, or knocked off the mat

- **Visible cue:** the shirt or a sleeve falls off the mat, the mat is struck and shifts or bunches under the shirt, a finished fold comes apart in transit, or the two arms strike each other.
- **SOP rule broken:** Steps 1 to 6, keep the shirt on the mat on a clear path and release only after a settled placement.
- **Coaching note:** check the path and the landing spot before you move.

Violation: Wrong episode ending

- **Visible cue:** the episode ends with a snap still open, a sleeve showing, a layer sprung open, the collar covered or face down, the shirt still spread on the mat, or an arm away from home.
- **SOP rule broken:** Step 7, confirm the squared packet, the five closed snaps, the hidden sleeves, and the whole collar showing face up; return both arms home; then stop recording.
- **Coaching note:** confirm first. Homing is the last thing the arms do.

Failures that are not violations

These failures are not caused by how the task was run. Log them as system issues, discard the episode, and never use them for coaching.

- **Recording stopped or paused during the episode** (recording system).
- **Camera dropped frames or lost its feed** (capture system).
- **Hardware fault on an arm:** gripper failure, drift, controller caused collision, or motor error.
- **Defective snap:** will not click closed under three correct presses, pops open on its own while the shirt is untouched, or has a stud or socket pulling away from the placket. Replace the shirt before the next episode.
- **Shirt will not hold a fold:** stroked end to end and the fold still springs open every time. Press the shirt before the next episode.
- **Defective shirt:** found torn, stained, damp, with an open seam, or with a placket that will not lie flat.
- **Shirt is too large for the mat,** so it cannot lie spread inside the mat with clear mat all round, or a cuff sits outside an arm's reach when the shirt is staged.
- **Fold mat shifts out of place** or bunches under the shirt during the episode.

Annotation subtasks (from SOP)

1. Press one snap closed on the placket
2. Smooth the placket from collar to hem
3. Turn the shirt face down on the mat
4. Fold one side of the body in
5. Lay one sleeve flat behind the body

6. Fold the hem end up one third
7. Return both arms home and end the episode